Vaibhav Kundu

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For any queries please visit my website: portfolio-website or 3d-interactive-portfolio



Objective

Motivated B.Tech Computer Science student with practical experience in Machine Learning and Deep Learning. Currently a Research Trainee at DRDO (CABS), developing speech processing systems for tactical communications using the NVIDIA NeMo framework and models like MarbleNet and Titanet-L. Proficient in Python, TensorFlow, and PyTorch, with a track record of building predictive models and conducting comparative neural network studies to optimize model accuracy.

Education

Netaji Subhas Engineering College — Kolkata, India
B.Tech in Computer Science & Engineering; CGPA: 7.57 — *2022 – 2026*

South Point High School — Kolkata, India
Class XII (CBSE); Percentage: 70.4% — *2022*

South Point High School — Kolkata, India
Class X (CBSE); Percentage: 90.8% — *2020*

Technical Skills

Languages: Python, C, Core Java
Query Languages: SQL, PL/SQL
OS Tools: Windows, Linux
Machine Learning & AI: Natural Language Processing (NLP), TensorFlow, PyTorch, Scikit-Learn, NumPy, Matplotlib, Pandas, OpenCV
Web & App Development: Android Studio, Android Framework (Java), Django
Core Concepts & Tools: DSA, OS, Computer Networks
Natural Languages: English, Hindi, Bangla

Achievements & Honors

Best Paper Award – ICDMIS 2026 — Kolkata, India
Co-Author | Springer — *Oct 2026*

- Awarded Best Paper in Track 7 at the 3rd International Conference on Data Mining and Information Security (ICDMIS 2026) organized by Eminent College of Management and Technology and SLIIT.
- Co-authored research paper: **"Forensic DocStitch: Vision-Language Reconstruction of Fragmented English Documents for Cyber Forensics"**.

Best Paper Award – IEEE SPACE 2026 — Bengaluru, India
Co-Author | IEEE Aerospace & Electronic Systems Society (AESS) — *Jul 2026*

- Awarded Best Paper in the *Assembly, Integration & Testing / Checkout* track at the 3rd IEEE International Conference on Space, Aerospace & Defence (SPACE 2026).
- Co-authored research paper: **"AI-Based Voice Deception System for Aerospace and Defense"** in collaboration with DRDO (Centre for Airborne Systems).

Experience

DRDO, Centre for Airborne Systems (CABS) — Bengaluru, India
Research Trainee — *Aug 2025 – Feb 2026*

- Developing a Speaker Identification, Diarization, and Voice Conversion system using NVIDIA NeMo framework.
- Utilizing advanced models such as MarbleNet, Titanet-L, and MSDD for speech processing tasks.

Sasken Technologies — Remote
Android Development Intern — *Jun 2025 – Jul 2025*

- Built a School Management App using Java (Android Framework) with modules for student profiles, notices, queries, attendance, and leave tracking.
- Enhanced communication channels between students, parents, and administrators.

IEEE Computational Intelligence Society Kolkata Chapter — Remote
Summer Internship — *Jun 2025 – Jul 2025*

- Conducted a comparative study between the performances of CNN (Convolutional Neural Network) and ViT (Vision Transformer) on the same dataset.

Celebal Technologies — Remote
Summer Internship — *May 2025 – Jul 2025*

- Built a Python system to predict student exam scores using Linear Regression.
- Utilized Pandas/NumPy for data preparation, Matplotlib for visualization, and Scikit-learn for modeling.

Infosys Springboard Remote
Project Intern *Oct 2024 – Dec 2024*

- Built a machine learning model to detect patient diseases based on symptoms and biological parameters.
- Tech Stack Used: Machine Learning, Python, Django.

Major Projects

AI-Based Voice Deception System Bengaluru, India
Research Project | DRDO (CABS) *Aug 2025 – Feb 2026*

- Engineered a Speaker Identification, Diarization, and Voice Conversion system for aerospace and defense applications using the NVIDIA NeMo framework.
- Co-authored the research paper "AI-Based Voice Deception System for Aerospace and Defense", which won the Best Paper Award at IEEE SPACE 2026.

Forensic DocStitch Kolkata, India
Final Year Project *2026*

- Engineered a vision-language reconstruction system capable of stitching and restoring fragmented English documents for cyber forensic applications.
- Co-authored a research paper detailing the methodology, which secured the Best Paper Award at the ICDMIS 2026 conference published by Springer.

Position of Responsibility

GNX – Netaji Subhash Engineering College Kolkata, India
Vice President *Apr 2025 – Dec 2025*

- Official GNU/Linux group and open-source community of NSEC.
- Previously served as Public Relations Team Member (Apr 2024 – Apr 2025).